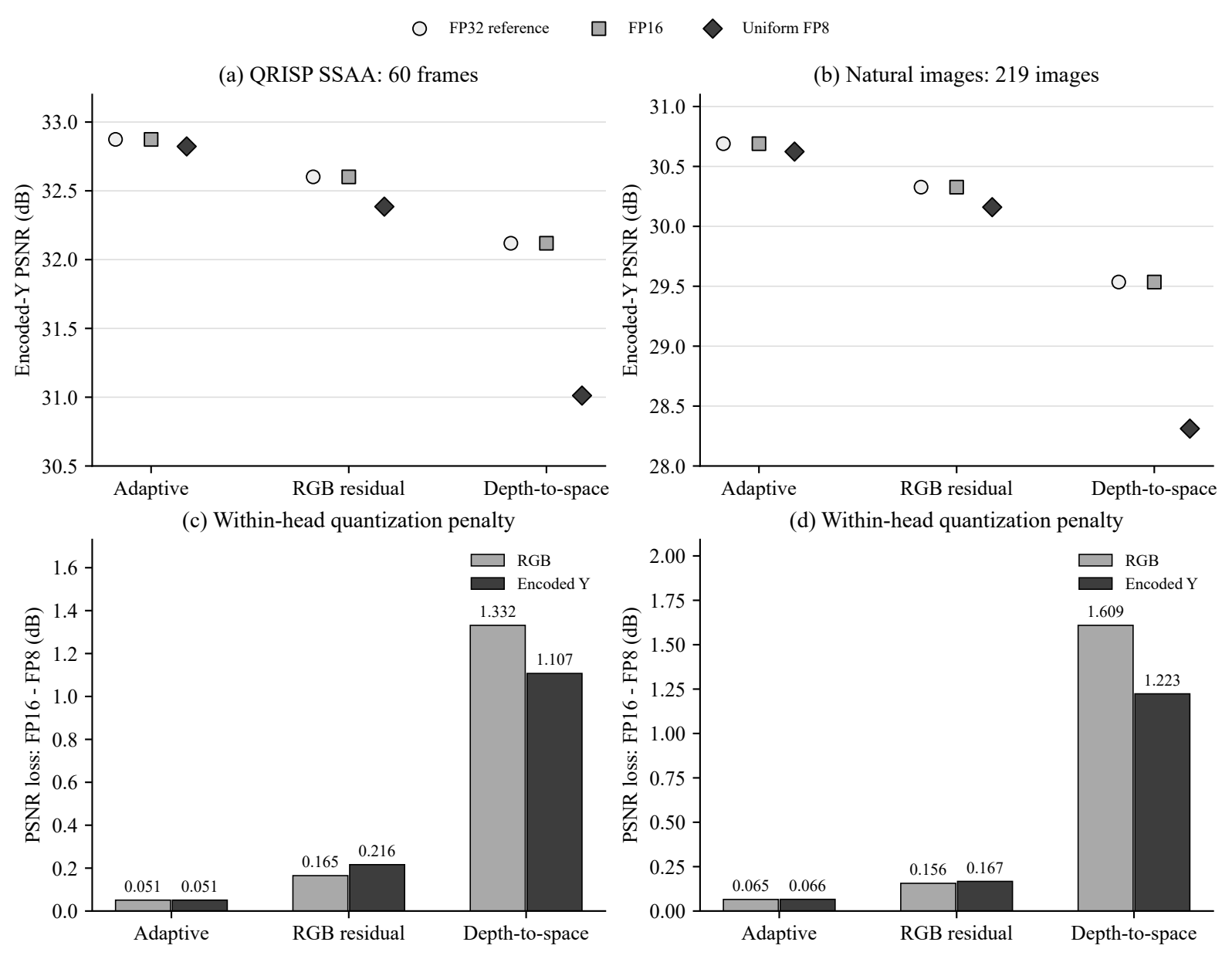
FP32 reference
FP16
Uniform FP8
(a) QRISP SSAA: 60 frames
Encoded-Y PSNR (dB)
33.0
32.5
32.0
31.5
31.0
30.5
Adaptive
RGB residual
Depth-to-space
(b) Natural images: 219 images
Encoded-Y PSNR (dB)
31.0
30.5
30.0
29.5
29.0
28.5
28.0
Adaptive
RGB residual
Depth-to-space
(c) Within-head quantization penalty
PSNR loss: FP16 - FP8 (dB)
RGB
Encoded Y
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.051
0.051
0.165
0.216
1.332
1.107
Adaptive
RGB residual
Depth-to-space
(d) Within-head quantization penalty
PSNR loss: FP16 - FP8 (dB)
RGB
Encoded Y
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.065
0.066
0.156
0.167
1.609
1.223
Adaptive
RGB residual
Depth-to-space